Worksheet 4.7 — The Normal Distribution

Mathematics: Analysis & Approaches SL (IB) · Unit 4: Statistics and Probability

The Big Ideas

Z-scores standardize any normal distribution, letting probabilities and percentiles be found (or reversed).

- $z = \dfrac{x - \mu}{\sigma}$.
- Inverse normal: given a probability, work backward for $x = \mu + z\sigma$.

Worked Examples

Example 1 — Z-score and probability

$\mu = 80$, $\sigma = 6$. Find $P(X < 89)$ given $P(Z < 1.5) \approx 0.9332$.

Solution. $z = \dfrac{89 - 80}{6} = 1.5$.

Conclusion: $P(X < 89) \approx 0.9332$.

Example 2 — Inverse normal

$\mu = 80$, $\sigma = 6$. Find the score at the 90th percentile, given $P(Z < 1.2816) \approx 0.90$.

Solution. $x = 80 + 1.2816(6)$.

Conclusion: $x \approx 87.69$.

Example 3 — Between two values

$\mu = 80$, $\sigma = 6$. Find $P(71 < X < 89)$ given $P(Z < -1.5) \approx 0.0668$ and $P(Z < 1.5) \approx 0.9332$.

Solution. $z_1 = -1.5$, $z_2 = 1.5$.

Conclusion: $P \approx 0.9332 - 0.0668 = 0.8664$.

Example 4 — Solve for mu and sigma

It's known $P(X < 60) = 0.1587$ and $P(X < 100) = 0.8413$. Find μ, σ (using $P(Z < -1) \approx 0.1587$, $P(Z < 1) \approx 0.8413$).

Solution. $\mu - \sigma = 60$, $\mu + \sigma = 100$. Add: $2\mu = 160 \Rightarrow \mu = 80$; $\sigma = 20$.

Conclusion: $\mu = 80$, $\sigma = 20$.

Example 5 — Finding P(X > a)

$\mu = 100$, $\sigma = 12$. Find $P(X > 121)$ given $P(Z < 1.75) \approx 0.9599$.

Solution. $z = \dfrac{121 - 100}{12} = 1.75$. $P(X > 121) = 1 - P(X < 121)$.

Conclusion: $P(X > 121) \approx 1 - 0.9599 = 0.0401$.

Example 6 — Finding a lower percentile

$\mu = 65$, $\sigma = 9$. Find the value below which the lowest 15% falls, given $P(Z < -1.0364) \approx 0.15$.

Solution. $x = 65 + (-1.0364)(9)$.

Conclusion: $x \approx 55.67$.

Example 7 — Sample proportion within a bound

$\mu = 500$, $\sigma = 25$. A batch is rejected if a measurement is outside $[450, 550]$. Find the proportion rejected, given $P(Z < -2) \approx 0.0228$ and $P(Z < 2) \approx 0.9772$.

Solution. $z_1 = \dfrac{450 - 500}{25} = -2$, $z_2 = \dfrac{550 - 500}{25} = 2$. Accepted proportion: $0.9772 - 0.0228 = 0.9544$.

Conclusion: rejected proportion $\approx 1 - 0.9544 = 0.0456$ (4.56%).

Example 8 — Solving with an asymmetric pair of z-values

It's known $P(X < 70) = 0.0668$ and $P(X < 130) = 0.9938$ for a normal distribution. Find μ, σ, using $P(Z < -1.5) \approx 0.0668$ and $P(Z < 2.5) \approx 0.9938$.

Solution. $\mu - 1.5\sigma = 70$ and $\mu + 2.5\sigma = 130$. Subtract: $4\sigma = 60 \Rightarrow \sigma = 15$.

Conclusion: $\sigma = 15$; $\mu = 70 + 1.5(15) = 92.5$.

Example 9 — Application — setting a pass threshold

A company wants only the top 10% of test scores (normal, $\mu = 72$, $\sigma = 8$) to qualify for advancement. Find the minimum qualifying score, given $P(Z < 1.2816) \approx 0.90$.

Solution. The top 10% corresponds to the 90th percentile: $x = 72 + 1.2816(8)$.

Conclusion: minimum score ≈ 82.25, so a score of 83 or higher would qualify.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

$\mu = 45$, $\sigma = 4$. Find the z-score for $x = 53$.

Question 2

$\mu = 120$, $\sigma = 10$. Find $P(X < 135)$ given $P(Z < 1.5) \approx 0.9332$.

Question 3

$\mu = 120$, $\sigma = 10$. Find the score at the 95th percentile, given $P(Z < 1.645) \approx 0.95$.

Question 4

$\mu = 120$, $\sigma = 10$. Find $P(105 < X < 135)$ given $P(Z < -1.5) \approx 0.0668$ and $P(Z < 1.5) \approx 0.9332$.

Question 5 — Challenge

A machine's fill weights are normal with unknown μ, σ; $P(X < 480) = 0.1587$, $P(X < 540) = 0.8413$. Find μ, σ.

Question 6

$\mu = 50$, $\sigma = 8$. Find the weight below which the lightest 10% falls, given $P(Z < -1.2816) \approx 0.10$.

Question 7

Explain why $P(X < \mu) = 0.5$ for any normal distribution.

Question 8

$\mu = 90$, $\sigma = 14$. Find $P(X > 115)$ given $P(Z < 1.786) \approx 0.9629$.

Question 9

$\mu = 200$, $\sigma = 30$. A batch is rejected outside $[140, 260]$; find the rejected proportion, given $P(Z < -2) \approx 0.0228$, $P(Z < 2) \approx 0.9772$.

Question 10

$P(X < 40) = 0.0668$ and $P(X < 100) = 0.9938$ for a normal distribution; find μ, σ (using $z = -1.5$ and $z = 2.5$).

Question 11 — Challenge

A company wants the top 5% of scores (normal, $\mu = 68$, $\sigma = 10$) to qualify; find the minimum qualifying score, given $P(Z < 1.645) \approx 0.95$.

Question 12 — Challenge

$\mu = 300$, $\sigma = 20$. Find the value below which the lowest 5% falls, given $P(Z < -1.645) \approx 0.05$.

Question 13 — Challenge

It's known $P(X < 55) = 0.0228$ and $P(X < 115) = 0.9938$; find μ, σ (using $z = -2$ and $z = 2.5$).

Question Recap & Answer Key

Question Recap

1. $\mu = 45$, $\sigma = 4$. Find the z-score for $x = 53$.
2. $\mu = 120$, $\sigma = 10$. Find $P(X < 135)$ given $P(Z < 1.5) \approx 0.9332$.
3. $\mu = 120$, $\sigma = 10$. Find the score at the 95th percentile, given $P(Z < 1.645) \approx 0.95$.
4. $\mu = 120$, $\sigma = 10$. Find $P(105 < X < 135)$ given $P(Z < -1.5) \approx 0.0668$ and $P(Z < 1.5) \approx 0.9332$.
5. A machine's fill weights are normal with unknown μ, σ; $P(X < 480) = 0.1587$, $P(X < 540) = 0.8413$. Find μ, σ.
6. $\mu = 50$, $\sigma = 8$. Find the weight below which the lightest 10% falls, given $P(Z < -1.2816) \approx 0.10$.
7. Explain why $P(X < \mu) = 0.5$ for any normal distribution.
8. $\mu = 90$, $\sigma = 14$. Find $P(X > 115)$ given $P(Z < 1.786) \approx 0.9629$.
9. $\mu = 200$, $\sigma = 30$. A batch is rejected outside $[140, 260]$; find the rejected proportion, given $P(Z < -2) \approx 0.0228$, $P(Z < 2) \approx 0.9772$.
10. $P(X < 40) = 0.0668$ and $P(X < 100) = 0.9938$ for a normal distribution; find μ, σ (using $z = -1.5$ and $z = 2.5$).
11. A company wants the top 5% of scores (normal, $\mu = 68$, $\sigma = 10$) to qualify; find the minimum qualifying score, given $P(Z < 1.645) \approx 0.95$.
12. $\mu = 300$, $\sigma = 20$. Find the value below which the lowest 5% falls, given $P(Z < -1.645) \approx 0.05$.
13. It's known $P(X < 55) = 0.0228$ and $P(X < 115) = 0.9938$; find μ, σ (using $z = -2$ and $z = 2.5$).

✓ Answer Key

1. $z = 2$
2. $P \approx 0.9332$
3. $x = 120 + 1.645(10) \approx 136.45$
4. $0.9332 - 0.0668 = 0.8664$
5. $\mu - \sigma = 480$, $\mu + \sigma = 540$: $\mu = 510$, $\sigma = 30$
6. $50 + (-1.2816)(8) \approx 39.75$
7. the normal distribution is symmetric about μ, so exactly half the probability lies below the mean
8. $z = (115 - 90)/14 \approx 1.786$; $P(X > 115) \approx 1 - 0.9629 = 0.0371$
9. $z_1 = -2, z_2 = 2$; accepted $= 0.9772 - 0.0228 = 0.9544$; rejected ≈ 0.0456
10. $\mu - 1.5\sigma = 40$, $\mu + 2.5\sigma = 100$; subtract: $4\sigma = 60 \Rightarrow \sigma = 15$; $\mu = 40 + 1.5(15) = 62.5$
11. $68 + 1.645(10) \approx 84.45$, so a score of 85 or higher qualifies
12. $300 + (-1.645)(20) \approx 267.1$
13. $\mu - 2\sigma = 55$, $\mu + 2.5\sigma = 115$; subtract: $4.5\sigma = 60 \Rightarrow \sigma \approx 13.33$; $\mu = 55 + 2(13.33) \approx 81.67$